

Problem 1

Solution:

(a) We'll show the reverse case first, i.e. $x = 0 \implies |x| = 0$. Since $x \geq 0$, $x = |x|$ so it follows that $|x| = 0$. Now, for the forward case, we must show that $|x| = 0 \implies x = 0$. To do this, we'll prove the contrapositive, i.e. $x \neq 0 \implies |x| \neq 0$. There are two cases: $x > 0$ and $x < 0$. If $x > 0$, then $|x| = x > 0$. And if $x < 0$ then $|x| = -x > 0$. In both cases, $|x| \neq 0$.

(b) We have 2 cases, $x > 0$ and $x \leq 0$. If $x > 0$ then $|x| = x$. However, we also have $-x < 0$, which means that $|-x| = -(-x) = x$. Thus, $|x| = |-x|$. Now, if $x \leq 0$, then $-x \geq 0$ so $|x| = -x$ and $|-x| = -x$. So yet again we get $|x| = |-x|$.

(c) Let $k = y - x$. From part (b), we have $|-k| = |k|$. Hence, by substitution, $|(y - x)| = |y - x| \implies |x - y| = |y - x|$

(d) We have 2 cases, $x \geq 0$ and $x < 0$. If $x \geq 0$ then $|x| = x \implies |x|^2 = x^2$. If $x < 0$, then $|x| = -x \implies |x|^2 = (-x)(-x) = (x)(x) = x^2$. Thus, in all cases $|x|^2 = x^2$.

(e) We have 2 cases, $x \geq 0$ and $x < 0$. If $x \geq 0$ then $|x| = x = \sqrt{x^2}$. If $x < 0$ then $|x| = -x = \sqrt{(-x)^2} = \sqrt{x^2}$. Thus, in all cases $|x| = \sqrt{x^2}$.

(f)

$$\begin{aligned} x \geq 0 &\implies |x| = x \\ y \geq 0 &\implies |y| = y \\ x \geq 0 \wedge y \geq 0 &\implies xy \geq 0 \implies |xy| = xy \\ x \geq 0 \wedge y \geq 0 &\implies |xy| = xy = |x|y = |x||y| \end{aligned}$$

$$\begin{aligned} x < 0 &\implies -x > 0 \implies |x| = -x = |-x| \\ y \geq 0 &\implies |y| = y \\ x < 0 \wedge y \geq 0 &\implies (-x)y > 0 \implies (-x)y = |(-x)y| = |-xy| = |xy| \\ x < 0 \wedge y \geq 0 &\implies |xy| = (-x)y = |x|y = |x||y| \end{aligned}$$

Without loss of generality, the above proof covers the $y < 0$ case as well.

$$\begin{aligned} x < 0 &\implies -x > 0 \implies |x| = -x = |-x| \\ y < 0 &\implies -y > 0 \implies |y| = -y = |-y| \\ x < 0 \wedge y < 0 &\implies (-x)(-y) = xy > 0 \implies |(-x)(-y)| = (-x)(-y) = xy = |xy| \\ x < 0 \wedge y < 0 &\implies |xy| = (-x)(-y) = |x|(-y) = |x||y| \end{aligned}$$

In all cases, we have $|xy| = |x||y|$

(g) Lemma: $|\frac{1}{y}| = \frac{1}{|y|}$.

Proof: observe that

$$\begin{aligned}
 1 &= |1| && \text{(since } 1 > 0\text{)} \\
 &= |y \frac{1}{y}| && \text{(for any real } y \neq 0\text{)} \\
 &= |y| |\frac{1}{y}| && \text{(by part (f))} \\
 \implies |\frac{1}{y}| &= \frac{1}{|y|}
 \end{aligned}$$

Now, observe that

$$|\frac{x}{y}| = |x \frac{1}{y}| = |x| |\frac{1}{y}| \quad \text{(by part (f))} = |x| \frac{1}{|y|} \quad \text{(by lemma)} = \frac{|x|}{|y|}$$

(h) The triangle inequality states $|a + b| \leq |a| + |b|$ for all real a, b . If we take $a = x$ and $b = -y$, we get $|x - y| \leq |x| + |-y|$. But from part (b) we know that $|-y| = |y|$ so this simplifies to $|x - y| \leq |x| + |y|$.

(i) The triangle inequality states $|a + b| \leq |a| + |b|$ for all real a, b . If we take $a = (x - y)$ and $b = y$, we get $|(x - y) + y| \leq |x - y| + |y|$. The lefthand side simplifies to $|x|$ so we have $|x| \leq |x - y| + |y|$. This in turn implies $|x| - |y| \leq |x - y|$.

(j) From the definition of absolute value, we have two cases: either $||x| - |y|| = |x| - |y|$ or $||x| - |y|| = -(|x| - |y|) = |y| - |x|$. By part (i) we know that $|x| - |y| \leq |x - y|$ and, similarly, $|y| - |x| \leq |y - x|$. By part (c), we have $|y - x| = |x - y|$. Thus, in both cases we have $||x| - |y|| \leq |x - y|$. $\square$

Problem 2

Solution:

Only including the calculations for a_6 and a_8 as the rest are similar.

For a_6 , since $|x + 2| \geq 5$ we have two cases: either $|x + 2| = 5$ or $|x + 2| > 5$. We'll start with the former. From the absolute value definition, either $5 = |x + 2| = x + 2$ or $5 = |x + 2| = -(x + 2)$. The first case implies that $x = 3$ is a valid solution. The second implies that $x = -7$ works as well. For $|x + 2| > 5$, observe that by the contrapositive to theorem 1.38 $x < -a \vee x > a \iff |x| > a$. Hence, either $x + 2 > 5$ or $x + 2 < -5$. These simplify to $x > 3$ or $x < -7$. Thus, we have $x = 3 \vee x = -7 \vee x > 3 \vee x < -7$, which we can simplify to $x \leq -7 \vee x \geq 3$. This is precisely b_8 .

For a_8 , since $|x + 5| < |x + 1|$ by theorem 1.38 we have $-|x + 1| < x - 5 < |x + 1|$. We'll start with the righthand inequality. By the same reasoning as in a_6 above, we have either $x + 1 > x - 5$ or $x + 1 < -(x - 5) = -x + 5$. Observe that in the first case, any real x works since $1 > -5 \implies 1 + a > -5 + a$ for all real a . In the second case $2x < 4 \implies x < 2$. Now, we can move to the lefthand inequality. $-|x + 1| < x - 5 \implies |x + 1| > -x + 5$, which gives us two cases: either $x + 1 > -x + 5$ or $x + 1 < -(-x + 5) = x - 5$. In the first case, we get $x > 2$. In the second, no real x works since $1 > -5 \implies 1 + a > -5 + a$ for all real a . Hence, we have the following cumulative condition on x . $[\text{any } x \in \mathbb{R} \vee x < 2] \wedge [x > 2]$, which is equivalent to $x > 2$. This is precisely b_4 .

- a_1, b_2
- a_2, b_5
- a_3, b_7
- a_4, b_{10}
- a_5, b_3
- a_6, b_8
- a_7, b_9
- a_8, b_4
- a_9, b_6
- a_{10}, b_1

□

Problem 3

Solution:

- (a). False. Let $a = -10$. Then, by defn of absolute value $|a| = -a = -(-10) = 10 \not\leq 5$.
- (b) True. By theorem 1.38 we have $-2 < x - 5 < 2$ which implies $3 < x < 7$
- (c) True. By theorem 1.38 we have $-1 \leq 1 + 3x \leq 1$ which implies $-2 \leq 3x \leq 0$. The first half implies $-\frac{2}{3} \leq x$
- (d) False. Let $x = \frac{3}{2}$. Then, $x - 1 = \frac{1}{2} > 0$ and $x - 2 = -\frac{1}{2} < 0$ so $|x - 1| = x - 1 = \frac{1}{2}$ and $|x - 2| = -(x - 2) = -(-\frac{1}{2}) = \frac{1}{2}$. Thus, $|x - 1| = |x - 2|$.
- (e) False. Let $x = 3$. Then, we have $|6 + y| = 5$. By the defn. of absolute value, either $|6 + y| = 6 + y = 5$ or $|6 + y| = -(6 + y) = 5$. In the first case $y = -1$. In the second case $6 + y = -5$ so $y = -11$. Thus, there is no positive y that satisfies the conditions. □

Problem 4

Solution:

We need to show that

$$\left(\sum_{k=1}^n a_k b_k\right)^2 = \left(\sum_{k=1}^n a_k^2\right) \left(\sum_{k=1}^n b_k^2\right) \iff \exists x, \forall k \ a_k x + b_k = 0$$

We'll start with the reverse case. Suppose that

$$\begin{aligned} a_1 x + b_1 &= 0 \\ a_2 x + b_2 &= 0 \\ &\vdots \\ a_n x + b_n &= 0 \end{aligned}$$

Then, we have, for all k $b_k = -a_k x$ and $a_k x = -b_k$. Taking the first result, we have

$$\left(\sum_{k=1}^n a_k b_k\right)^2 = \left(\sum_{k=1}^n a_k (-a_k x)\right)^2 = \left(-x \sum_{k=1}^n a_k^2\right)^2 = x^2 \left(\sum_{k=1}^n a_k^2\right)^2$$

Now, taking the second result we have $a_k x = -b_k \implies a_k^2 x^2 = (-b_k)^2 = b_k^2$.

Summing over all k gives us

$$\begin{aligned} a_1^2 x^2 + a_2^2 x^2 + \cdots + a_n^2 x^2 &= b_1^2 + b_2^2 + \cdots + b_n^2 \\ x^2 (a_1^2 + a_2^2 + \cdots + a_n^2) &= b_1^2 + b_2^2 + \cdots + b_n^2 \\ x^2 \left(\sum_{k=1}^n a_k^2\right) &= \sum_{k=1}^n b_k^2 \end{aligned}$$

Thus, we have

$$\left(\sum_{k=1}^n a_k b_k\right)^2 = x^2 \left(\sum_{k=1}^n a_k^2\right)^2 = \left(\sum_{k=1}^n a_k^2\right) \left[x^2 \left(\sum_{k=1}^n a_k^2\right) \right] = \left(\sum_{k=1}^n a_k^2\right) \left(\sum_{k=1}^n b_k^2\right)$$

Now we'll move to the forward case. Suppose

$$\left(\sum_{k=1}^n a_k b_k\right)^2 = \left(\sum_{k=1}^n a_k^2\right) \left(\sum_{k=1}^n b_k^2\right)$$

Following Apostol's proof of the inequality, let

$$\begin{aligned} A &= \sum_{k=1}^n a_k^2 \\ B &= \sum_{k=1}^n a_k b_k \\ C &= \sum_{k=1}^n b_k^2 \end{aligned}$$

so that our assumption reads $B^2 = AC$. For every real x ,

$$\sum_{k=1}^n (a_k x + b_k)^2 = \sum_{k=1}^n (a_k^2 x^2 + 2a_k b_k x + b_k^2) = Ax^2 + 2Bx + C$$

We have 2 cases: $A \neq 0$ and $A = 0$.

If $A \neq 0$, let $x = -\frac{B}{A}$. Then

$$\begin{aligned} Ax^2 + 2Bx + C &= A \left(\frac{B^2}{A^2} \right) - \frac{2B^2}{A} + C \\ &= \frac{B^2}{A} - \frac{2B^2}{A} + C \\ &= \frac{AC - B^2}{A} \\ &= 0 \end{aligned} \quad (\text{since } B^2 = AC)$$

so $\sum_{k=1}^n (a_k x + b_k)^2 = 0$. A sum of squares of real numbers is 0 only when every term is 0: each term is ≥ 0 , so if any one term were > 0 the whole sum would be > 0 . Hence $a_k x + b_k = 0$ for every k , with this single x .

If $A = 0$, then $\sum_{k=1}^n a_k^2 = 0$, and by the same sum-of-squares argument every $a_k = 0$. If every $b_k = 0$ as well, then any real x gives $a_k x + b_k = 0x + 0 = 0$ for every k .

If some $b_k \neq 0$, then $a_k x + b_k = 0x + b_k = b_k \neq 0$ for every real x , so no x exists and the statement as written fails. The statement as written fails here, so we swap the roles of a and b . Since every $a_k = 0$, taking $x = 0$ gives $b_k x + a_k = 0 + 0 = 0$ for every k .

So the precise statement is: equality holds in the Cauchy-Schwarz inequality if and only if there is a real x with $a_k x + b_k = 0$ for every k , or a real x with $b_k x + a_k = 0$ for every k . The reverse case above and the two cases here prove both directions.

□